

Feature-Level Fusion of GAN and Diffusion Models for Image Enhancement

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Abstract. “With Generative Adversarial Networks (GANs) and Diffusion Models emerging as two of the most influential techniques, generative models have transformed image synthesis, enhancement, and transformation tasks in recent years.” Although GANs are excellent at quickly inferring textures that are visually realistic, they are frequently constrained by problems such as mode collapse and training instability. However, Diffusion Models have slower generation times but produce high-fidelity outputs with stable training dynamics. In addition to examining new hybrid architectures for image enhancement that combine the complementary powers of GANs and diffusion models, this review looks into the advantages and disadvantages of both paradigms. The model ESRGAN and LDM were never combined for the image enhancement objective.

Keywords: *GAN, Diffusion Model, Image Blending, Image Super-Resolution, Latent Diffusion, ESRGAN, Image Enhancement.*

Introduction:

Image Enhancement is the process of improving the visual quality of a digital image to make it more useful for display or analysis by increasing contrast, removing noise, and highlighting details [6]. It is widely used for image enhancement. “A crucial task in computer vision and artificial intelligence is image enhancement, particularly the conversion of low-resolution (LR) images into their high-resolution (HR)” [2]. “High-quality image super-resolution is becoming more and more in demand.” “Stable diffusion models and Generative Adversarial Networks (GANs) are some of the most well-known generative models” [4], [11].



Fig. 1. Low Resolution Image



Fig. 2. Enhanced Image

A. Generative Adversarial Networks (GANs)

“GAN first introduced by Ian Goodfellow and his colleagues in 2014” [1]. “GAN has various applications like image generation, image enhancement” [1], [6]. “There are two main parts of the GAN the first is the Generator and the other is the Discriminator.” “Generators produce high quality images which mimic the actual image, and discriminators try to find out if the generated image is real or fake” [1]. “If the discriminator predicts the image is fake, then it is calculated as the loss of the generator and if not then it is the loss of the discriminator” [1]. “GAN has shown effective results in image enhancement.” “The ESRGAN is one of the examples of GAN, which is used for image super-resolution.” “By utilising Residual-in-Residual Dense Blocks (RRDBs), ESRGAN enhances previous models by enabling the generator to preserve finer structures and deeper information.” “Although GAN produces fast images, it faces issues like mode collapse.”

B. Diffusion Models

One kind of generative model is a diffusion model, which learns by gradually introducing noise into an image before denoising it [11]. “The forward diffusion phase and the reverse denoising phase are the two phases of the training process” [11]. “To efficiently capture intricate picture patterns and spatial correlations, diffusion models frequently employ a U-Net design” [3], [11]. “Diffusion models are able to overcome the limitations of GANs, such as mode collapse, because of their steady optimisation process and probabilistic training framework” [4], [9]. “Diffusion models provide detailed and diverse high-resolution images.” Diffusion Models result in images with remarkable detail and smooth transitions. Diffusion models are stable and perform better than GAN, but they are slower than GAN. To tackle the limitations of GAN and diffusion models, Hybrid models that combine GAN and Diffusion models are good options.

Literature Review:

“Xintao Wang et al. (2018) [1] proposed ESRGAN (Enhanced Super-Resolution Generative Adversarial Network), an improved version of SRGAN that incorporates Residual-in-Residual Dense Blocks (RRDBs) to strengthen feature representation.” “This architecture allows ESRGAN to create new, improved images.”

“Bee Lim et al. (2017) [2] show the ESDR network, which used the DIV2K dataset for training, ESDR performed and showed the reconstruction of the image and set the new benchmarks for image enhancement”

Robin Rombach et al. (2022) [3] proposed Latent Diffusion Models (LDMs), in which LDMs work on the latent space, converting the image into latent space. It is low-cost and perform well for texturing the image.”

“Prafulla Dhariwal and Alex Nichol (2021) [4] introduced Denoising Diffusion Probabilistic Models (DDPMs). They generate high quality image using the reverse time stochastic method.” The results outperformed other Generative models.

Minguk Kang et al. (2025) [5] proposed a hybrid framework that combines DM and Conditional GAN the goal was to combine models to result in a high-quality image. They used large datasets, like COCO and LAION.

Christian Ledig et al. (2017) [6] introduced SRGAN, a super-resolution model to leverage loss within the GAN. This model calculate pixel wise loss. It suppresses the other image enhancement methods.

Yulun Zhang et al. (2018) [7] proposed the Residual Channel Attention Network (RCAN) for image improvement. It enhanced reconstruction performance with the CNN-based models by integrating channel attention mechanisms into deep residual learning

Sanchayan Vivekananthan (2025) [8] evaluated Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Diffusion Models across the whole image for image enhancement. The study shows the strengths and limitations of the GAN and DM.

Yingying Peng (2025) [9] completed a study of GAN and DM for image generation with the CIFAR-10 dataset and the ImageNet dataset. the Study conduction hybrid approach it effectively works and balances both models

Xuechao Zou et al. (2023) [10] proposed the conditional diffusion model for cloud removal from satellite images. The result of the model effectively saved the details of the image and its structure, of the image and gave clarity. The DM works well with geological datasets.

Jonathan Ho et al. (2020) [11] laid the mathematical foundation for diffusion-based generation by formally introducing the DDPM framework. The model gradually adds and then removes noise through a diffusion process, producing sharp and realistic images. This seminal work sparked a wave of research into diffusion-based generative models.

Research Gap:

A. Limitations of ESRGAN and LDM

Mode collapse problem in GANs, GANs are fast, but face mode collapse issues; diffusion models overcome this limitation.

B. Integration of ESRGAN and LDM for Image Enhancement

In previous studies, the focus was only on either ESRGAN or on the LDM. There is no particular study that focuses on combining these two models for the image enhancement objective.

Proposed Methodology:

A. Dataset Details

Source: The dataset used in this study is the MetFace dataset, obtained from Kaggle. The dataset contains high-resolution, diverse human face images. The dataset is designed to improve realism and diversity.

The MetFace dataset is suitable for evaluating super-resolution and image enhancement performance since it provides high-resolution images with a range of textures and structural details, despite its emphasis on human faces. However, the application of this study to non-facial areas remains a problem. The dataset contains 1336 high-quality face images.

Attribute	Description
Name	MetFace
Total Images	1336
Image Resolution	512x512

Table 1. Data set Description



Fig. 3. Dataset Sample Images

B. System Architecture

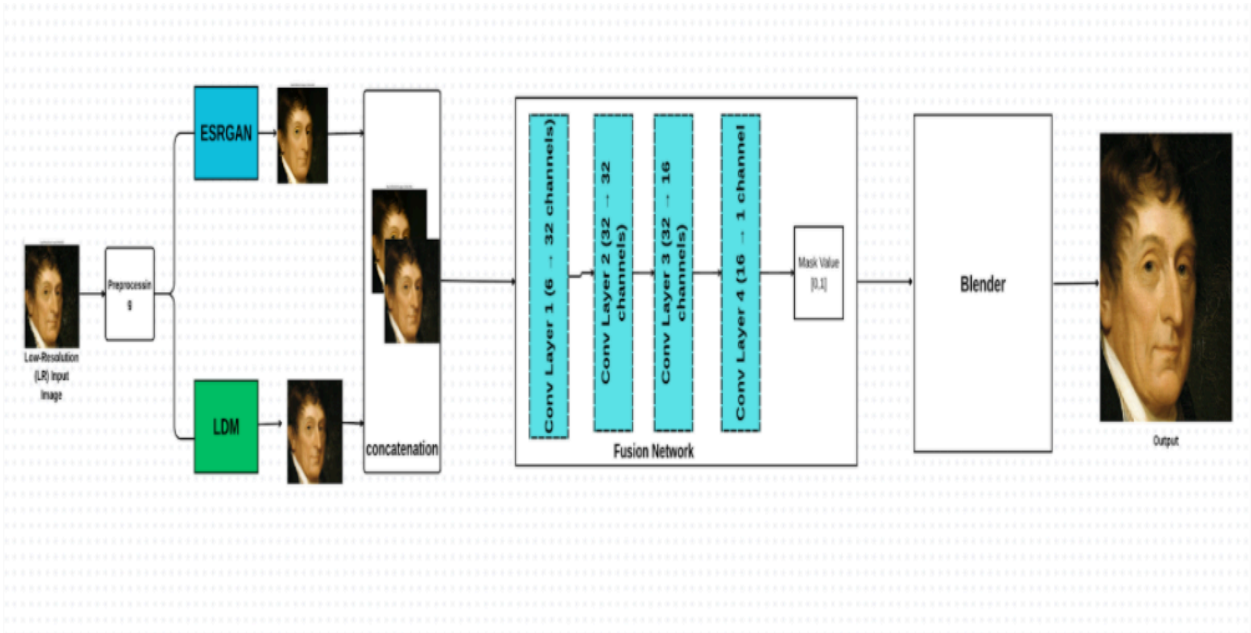


Fig. 4. System Architecture

C. Preprocessing:

Cropping: Cropping was performed to eliminate redundant background information and to retain only the region of interest, thereby improving feature relevance and reducing computational complexity.

Resizing: To ensure uniform input dimensions for the model, all photos were downsized to a constant resolution while maintaining the original aspect ratio to avoid geometric distortion.

D. ESRGAN and LDM :

In the next step, the same input is passed to the ESRGAN and LDM at the same time; the models used for this are the pretrained models. Output is generated by both models and passes for the concatenation.

E. Concatenation :

In Concatenation, using the PyTorch library, both outputs from ESRGAN and LDM are combined with each other. For the image, we have 3RGB values for one pixel, but after concatenation, we will have 6RGB values for one pixel, 3RGB values of ESRGAN and 3RGB values of LDM.

F. Fusion Network:

The fusion network is designed to integrate the complementary outputs of the GAN-based and diffusion-based generative branches. Feature maps generated by both models are first spatially aligned and concatenated along the channel dimension, forming a unified representation that combines global structure and fine-grained details.

“The concatenated feature tensor is processed through a sequence of convolutional layers.” “Specifically, the fusion network consists of four convolutional blocks with kernel size 3×3 .” “The first block employs 64 filters, followed by blocks with 128, 64, and 32 filters, respectively.” “Each convolutional layer is followed by a ReLU activation

function to add non-linearity and enhance feature discrimination, as well as batch normalisation to increase training stability and convergence speed.” “A final 1×1 convolution with 3 filters is applied to map the fused features to the output image space.” The output layer uses a sigmoid activation function for smooth normalising the range and to get the mask value M .

The fusion network learns the combination of GAN and LDM-generated details while keeping the consistency of the structure.

G. Blender:

The blender combines the output of the fusion network using the blending image formula. The blender task is to calculate the pixel value based on the mask value for each pixel. Blender calculates the pixel value it. blender generate the two one pixel from the mask value. The mask value is calculated using the sigmoid activation function from the fusion network. The value of Mask M is between 0 and 1. The result we get from the sigmoid activation function is always between 0 and 1. The formula for blending images is

$$\text{Fused}([RGB]) = M * \text{ESRGAN}([RGB]) + (1 - M) * \text{LDM}([RGB]) \quad (1)$$

The mask value reflects the details while keeping the global structure safe and consistent. This blender allows for a smooth transition of pixels in the image. It removes the noise and reduces the artefacts for the image; the blender improves the structural and perceptual quality of the end output.

The adaptive blender further refines the output by giving each model spatially variable weights, while the fusion network is completely trainable and learns optimal feature interactions.

Training and Loss Functions: The pretrained ESRGAN and LDM parameters are fixed while the fusion network and blender undergo end-to-end training. The training goal combines a perceptual loss to improve visual realism and a reconstruction loss to maintain pixel-level accuracy, guaranteeing both structural consistency and fine-grained detail preservation.

Output:



Fig. 5. LDM, ESRGAN and fused outputs.

The output of the blender is used to create the final synthesised image. Several loss functions, including

Reconstruction and perceptual losses are used to enhance the overall framework and enable high-fidelity image generation.

Result:

The performance of the model is presented in this section, PSNR peak signal to noise ration main parameter used for the comparison and evaluation of the performance of the models. PSNR is calculated individually for ESRGAN, LDM and the final output after the fusion network and after blending the image. This result analysis shows how well the fusion network is performing along with the other two models, and reconstructing the final output. The result shows the improvement of the fusion network. The formula for PSNR is as below

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (2)$$

Model Output	PSNR
Real-ESRGAN vs. HR (512×512)	25.3594
LDM vs. HR (512×512)	26.5598
Fused Output vs. HR (512×512)	27.7862

Table 2. Results

The result shows 4.6% with LDM and With ESRGAN 9.5% improvement with ESRGAN when comparing the PSNR of models. This gives the idea that the fusion framework is successfully combining the power of both models.

Result Discussion:

The output of ESRGAN and its PSNR is 25.36 dB. It shows the model is working well, but failing to preserve small details. ESRGAN has the super-resolution ability but lacks in keeping small details. The PSNR of the Latent diffusion Model(LDM) is 26.5598dB, which shows LDM is working better than the ESRGAN. With PSNR, the higher result, the higher the accuracy. LDM enhance image while keeping details and textures safe. But the output of the fusion network outperforms both the models; the PSNR of the fusion network result is 27.79 dB, which is greater than both models, LDM and ESRGAN. This shows that the fusion models effectively combine the advantages of the LDM and ESRGAN and limit both models' limitations. Fusion network suppresses further improvement and improves structural consistency. Both models offer complementary information and detail for the final output, which the fusion network successful keeping it and combines the outputs. Overall, it confirms that blending at the feature level produces the high-resolution output.

Conclusion:

This study introduces the fusion network, which combines the strengths of both LDM and ESRGAN models. The suggested model successfully applied the advantages of both models and gives the mechanism for combines output of ESRGAB and LDM. The model successfully enhances the image from a low-resolution image while maintaining the details and structure of the image. The three models are evaluated using the PSNR. The result with PSNR, we can see the fusion model outperforms the ESRGAN and LDM. The final output shows the incense in visual clarity, with higher reconstruction of the image. By considering all things, the suggested fusion network offers a reliable solution for image enhancement and high-quality image reconstruction applications. In addition different perceptual metrics, large datasets, and other fusion techniques might be investigated in future research.

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